Wrench-Feasible Workspace Generation for Cable-Driven Robots

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***Abstract*—This paper presents a method for analytically generating the boundaries of the wrench-feasible workspace (WFW) for cable robots. This method uses the available net wrench set, which is the set of all wrenches that a cable robot can apply to its surroundings without violating tension limits in the cables. The geometric properties of this set permit calculation of the boundaries of the WFW for planar, spatial, and point-mass cable robots. Complete analytical expressions for the WFW boundaries are detailed for a planar cable robot and a spatial point-mass cable robot. The analytically determined boundaries are verified by comparison with numerical results. Based on this, several workspace properties are shown for point-mass cable robots. Finally, it is shown how this workspace-generation approach can be used to analytically formulate other workspaces.**

***Index Terms*—Cable robot, workspace generation, wrench-feasible, wrench set.**

I. Introduction

CABLE-DRIVEN robots, referred to as cable robots in this paper, are a type of robotic manipulator that has recently attracted interest for large workspace manipulation tasks. Cable robots are relatively simple in form, with multiple cables attached to a mobile platform or end-effector, as illustrated in Fig. 1. The end-effector is manipulated by motors that can extend or retract the cables. These motors may be in fixed locations or mounted to mobile bases. The end-effector may be equipped with various attachments, including hooks, cameras, electromagnets, and robotic grippers. Fig. 1 illustrates a cable robot with three cables equipped with a robotic gripping tool grasping a barrel.

These robots possess a number of desirable characteristics, including: 1) stationary heavy components and few moving parts, resulting in low inertial properties and high payload-to-weight ratios; 2) potentially vast workspaces, limited mostly by cable lengths, interference with surroundings, and force/moment exertion requirements; 3) transportability and ease of disassembly/reassembly; 4) reconfigurability by simply relocating the motors and updating the control system accordingly; and 5) economical construction and maintenance due to few moving parts and relatively simple components. Consequently, cable robots are exceptionally well-suited for many applications, such as manipulation of heavy payloads, haptics, cleanup of disaster sites, access to remote areas, and interaction with hazardous environments. A wide variety of cable robots have been developed (e.g., [1]–[5]). Fig. 2 shows the RoboCrane [1], a six-cable manipulator for use in tasks such as material handling and manufacturing operations. Fig. 3 shows the Cablecam [6], which is used to position a video camera in stadiums and arenas.

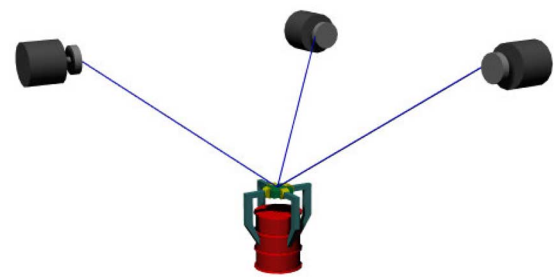

Fig. 1. Example of cable robot.

Fig. 2. Scale model of the RoboCrane [1].

One important class of cable robots are point-mass cable robots. In these manipulators, all cables attach to a single point on the end-effector, and can change lengths to control the position of the end-effector. Typically, the end-effector is modeled as a lumped mass located at the point of intersection of the cables. As an example, the manipulators in Figs. 1 and 3 can be modeled as point-mass cable robots. Because the structure of point-mass cable robots is simple, they are relatively easy to implement and are used in applications such as camera positioning [6], [7], haptics [3], [8], and cargo handling [2].

Manuscript received March 15, 2005; revised August 8, 2005. This paper was recommended for publication by Associate Editor R. Roberts and Editor I. Walker upon evaluation of the reviewers' comments. The work of P. Bosscher was supported by a National Defense Science and Engineering Graduate (NDSEG) Fellowship, and the work of A. T. Riechel and I. Ebert-Uphoff was supported in part by the National Science Foundation under Career Grant #CMS-9984279. This paper was presented in part at the IEEE International Conference on Robotics and Automation, New Orleans, LA, April 2004.

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Color versions of Figs. 1, 3–8, and 10–15 are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TRO.2006.878967

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Fig. 3. Cablecam [6].

Determining the usable workspaces of these cable robots is very important. Despite the fact that much work has been done in the area of generating workspaces of robotic manipulators (e.g., [9]–[11]), in most cases, these techniques cannot be used for cable robots, because cable robots have the limitation that the cables can pull, but not push, on the end-effector.

Many applications require the end-effector to operate in a space of a particular shape, and to exert certain minimum force/moment combinations (or *wrenches*) throughout that space. Accordingly the most appropriate workspace to consider is the *wrench-feasible workspace* (WFW). A pose of a cable robot is said to be *wrench-feasible* in a particular configuration and for a specified set of wrenches, if the tension forces in the cables can counteract any external wrench of the specified set applied to the end-effector [12]. The WFW is defined as the set of poses that are wrench-feasible. Thus, if a given set of wrenches must be exerted by the end-effector on its surroundings in order to accomplish a task, the manipulator can exert these required wrenches at any point in the corresponding WFW. This region, therefore, constitutes the workspace which is "usable" by the robot for a particular application.

While the WFW has been defined in general terms [12], [13], it has generally been formed numerically using an exhaustive search approach [5], [13]–[15]. One exception is [16], where the boundaries of the WFW were determined analytically for planar four-cable fully constrained[1] cable robots, assuming infinite upper tension limits. Some researchers have also incorporated workspace limits based on cable interference, but these workspace limits were determined either experimentally [5] or numerically [17].

Several researchers have investigated the set of all poses that the end-effector can attain statically (with no external forces or moments acting besides gravity) [1], [18]–[24], which is referred to here as the static equilibrium workspace (SEW). In most cases, formulation of the SEW has been done numerically via "brute force" methods, where the entire taskspace is discretized and exhaustively searched to find the statically reachable poses. Two exceptions are in [1] and [20], where the boundaries of the SEW were defined analytically, but both of these formulations relied on special manipulator geometries. The "dynamic workspace," which is defined as the set of all poses where the end-effector can be given a specific acceleration, has also been formulated analytically for planar cable robots [25].

[1]A cable robot is *underconstrained* if it relies on gravity to determine the pose of the end-effector, and is *fully constrained* if the pose of the end-effector is completely determined by the lengths of the cables.

The purpose of this paper is to provide a general analytical formulation of the WFW. This method applies to both underconstrained and fully constrained cable robots, and is also analytically based, thus the resulting description of the workspace provides insight into the workspace geometry that cannot be obtained through numerical approaches alone. The resulting contributions of this paper include the formulation of the boundaries of the WFW for planar, spatial, and point-mass cable robots, as well as analytical solutions for a number of example manipulators.

Organization: Section II reviews additional relevant literature. Section III presents a wrench-based analysis of general cable robots. This analysis is used in Section IV to develop a geometric interpretation of wrench feasibility, and an approach for generating the WFW by defining its boundaries analytically. Section V then applies the workspace-generation approach to general planar and spatial cable robots. Section VI describes the WFW boundary equations and resulting workspace properties for point-mass cable robots. Section VII discusses how other workspaces can be generated using the method developed here. Finally, Section VIII discusses and summarizes these results, and Section IX presents areas of future work.

II. Additional Related Work

In formulating the WFW and performing the disturbance robustness analysis, this paper uses a construction called the available net wrench set, the set of all forces and moments that the manipulator can exert without violating cable tension limits. Similar concepts have been developed by other researchers, including the "capable force region," which is defined in [26] as the set of forces that the manipulator can exert without consideration of the associated moments. In [27], a three-cable planar point-mass cable robot was examined. The set of all forces that the three cables could exert on the end-effector was termed the "set of manipulating forces." A similar set of wrenches was also defined in [25] and termed a "pseudopyramid." This pseudopyramid includes the set of all wrenches (force/moment combinations) that the cables could apply to the end-effector at a pose if the cables have no upper tension limits. In addition, the analysis of wrench feasibility presented here is similar to the analysis of disturbing and nondisturbing wrenches in [28].

III. Wrench-Set Analysis for General Cable Robots

In order to use a cable robot to accomplish desired tasks, the cables driving the end-effector must exert wrenches (force/moment combinations) on the end-effector. Given any considered pose (position and/or orientation) of the robot, it is possible to determine the set of all possible wrenches that the cables can apply to the end-effector, and thus, the set of all wrenches that the end-effector can apply to its surroundings.

In the analysis presented here, it is assumed that the cables have negligible mass and do not stretch or sag, the end-effector is a single rigid body with known cable attachment points on the end-effector relative to the center of gravity, the locations of the attachments of the cables to the motors are known, and each motor controls exactly one cable. Cable lengths, the direction of gravity, and the resulting pose of the mechanism are also assumed to be known.

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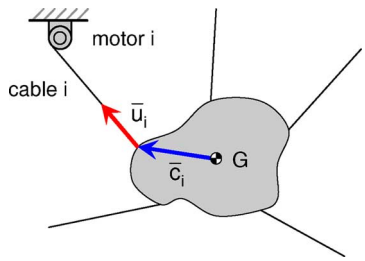

Fig. 4. Diagram of kinematic parameters.

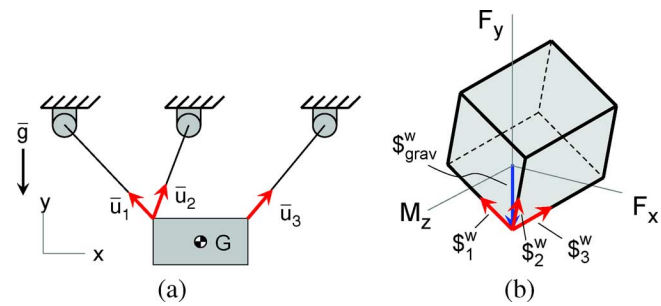

Fig. 5. Planar cable robot and its available net wrench set. (a) Example of planar cable robot. (b) Available net wrench set.

A. Available Net Wrench Set

Assuming positive tension in all cables, the Jacobian relationship for parallel robots holds for cable robots. Thus, the set of wrenches that can be applied to the end-effector can be formed by examining the positive range space of the transpose of the Jacobian matrix. This matrix describes the linear relationship between the cable tensions, $\overline{t} = \{t_1, \ldots, t_p\}^T$, and the resulting wrench $\$^w = \left(\frac{\overline{F}}{M}\right)$ at the end-effector. Let $\overline{u}_j$ be the unit vector running along cable i directed away from the end-effector, as shown in Fig. 4, $\overline{c}_i$ be the vector from G, the center of mass of the end-effector, to the point on the end-effector where cable i is connected, and let p cables be attached to the end-effector. The cable wrench set (CW) is then defined to be set of all force-moment combinations that can be generated at the center of gravity of the end-effector by the cables

$$\mathrm{CW} = \{\$^w : \$^w = \boldsymbol{J}^T\overline{t}, \quad t_{i,\max} \geq t_i \geq 0\} \tag{1}$$

where

$$\boldsymbol{J}^T = [\$^w_1, \ldots, \$^w_p] \tag{2}$$

and $\$^w_i$ is the wrench[2] along the ith cable

$$\$^w_i = \begin{Bmatrix} \overline{u}_i \\ \overline{c}_i \times \overline{u}_i \end{Bmatrix}. \tag{3}$$

The restriction that $t_{i,\max} \geq t_i \geq 0$ stems from the fact that each cable can pull, but not push (i.e., a cable cannot have negative tension), and is restricted to be less than or equal to a maximum tension $t_{i,\max}$. This maximum tension may be determined by the torque limits of the motor reeling in the cable or by the maximum tension a cable can withstand without breaking. Note that the definition in (1) holds for both redundant ($p > n$, where n is the dimension of the task space) and nonredundant ($p \leq n$) manipulators.

We now wish to form the set of wrenches that the end-effector can apply to its surroundings, taking into account the effect of constant external wrenches such as gravity. This set is termed the *available net wrench set*, abbreviated $\mathrm{NW}_{\mathrm{avail}}$. Assuming a constant external wrench $\left(\frac{\overline{F}}{M}\right)_{\mathrm{ext}}$ is applied to the end-effector (typically, the gravitational wrench $\$^w_{\mathrm{grav}} = \binom{m\overline{g}}{\overline{0}}$, where m is the mass of the end-effector and $\overline{g}$ is the gravitational vector, directed downward), the available net wrench set is then

$$\begin{aligned}\mathrm{NW}_{\mathrm{avail}} &= \mathrm{CW} \oplus \left(\frac{\overline{F}}{M}\right)_{\mathrm{ext}} \\ &= \left\{\$^w : \$^w = \boldsymbol{J}^T\overline{t} + \left(\frac{\overline{F}}{M}\right)_{\mathrm{ext}}, \quad t_{i,\max} \geq t_i \geq 0\right\}.\end{aligned} \tag{4}$$

B. Graphical Representation

If the dimension n of the task-space of the robot is less than or equal to three, it is possible to construct a graphical representation of $\mathrm{NW}_{\mathrm{avail}}$. As an example, consider the planar manipulator in Fig. 5(a). Given the geometry of the manipulator at the current pose, the unit vectors $\overline{u}_1, \overline{u}_2$, and $\overline{u}_3$ can be constructed. Applying (3) results in $\$^w_1$, $\$^w_2$, and $\$^w_3$. The set $\mathrm{NW}_{\mathrm{avail}}$ can then be expressed as $\mathrm{NW}_{\mathrm{avail}} = \{\$^w : \$^w = a_1 t_{1,\max}\$^w_1 + a_2 t_{2,\max}\$^w_2 + a_3 t_{3,\max}\$^w_3 + \$^w_{\mathrm{grav}}; 0 \leq a_i \leq 1\}$. Fig. 5(b) illustrates the resulting set, where $t_{\max}$ is assumed to be the same for all three cables. We can see here that $\mathrm{NW}_{\mathrm{avail}}$ is a parallelepiped. Note that this parallelepiped is defined in the mixed-dimensional space of $F_x - F_y - M_z$.

In general, it can be shown that $\mathrm{NW}_{\mathrm{avail}}$ is some form of a parallelogram, parallelepiped, or hyper-parallelepiped, depending on the number of cables and the dimension of the task-space. If p is the number of cables and q is the dimension of the task space ($q = 3$ in the planar and point-mass case, $q = 6$ in the spatial case), then $\mathrm{NW}_{\mathrm{avail}}$ has $2\binom{p}{q-1} = 2p!/(q-1)!(p-(q-1))!$ facets.

IV. WFW Generation

In many applications, the requirements for a task or set of tasks can be characterized by a required set of wrenches that the end-effector must apply to its surroundings. Given this requirement, the WFW is defined in [12] as the set of all poses that are wrench-feasible, i.e., where the manipulator can apply the required set of wrenches. Let this set of required wrenches be called $\mathrm{NW}_{\mathrm{req}}$, the *required net wrench set*. Then the WFW can be described as the set of all poses P of the end-effector, where

$$\mathrm{NW}_{\mathrm{req}}(P) \subseteq \mathrm{NW}_{\mathrm{avail}}(P). \tag{5}$$

[2]The notation of a wrench as $\$^w$ is used in order to remain consistent with the standard notation of screw theory. Note also that $\$^w_i$ does not actually have units of force and moment, but must be multiplied by a scalar force factor in order to take on the standard units of a wrench. $\$^w_i$ can also be thought of as simply a screw in ray coordinates.

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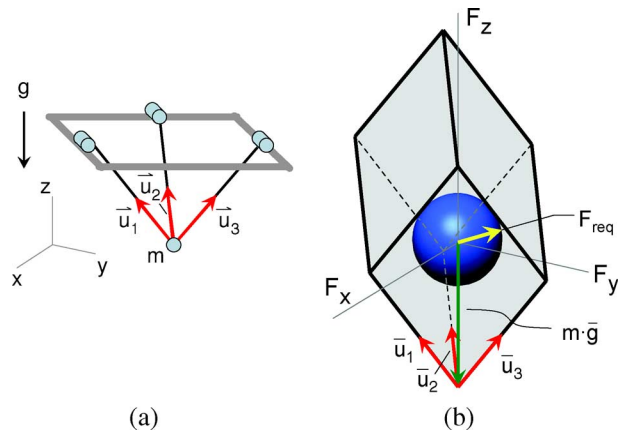

Fig. 6. Point-mass three-cable manipulator and its available net wrench set containing its required net wrench set. (a) Example manipulator. (b) Available net wrench set with required net wrench set.

Although $\mathrm{NW_{req}}$ can be chosen arbitrarily, it is typically chosen to be a geometrically simple set of wrenches that is independent of P (when expressed in a coordinate frame fixed to the end-effector).

For example, consider the point-mass three-cable manipulator shown in Fig. 6(a). Assume that the manipulator's task requires it to exert a force F_{req} in any direction. The corresponding choice for $\mathrm{NW_{req}}$ would then be the set of all forces $\overline{F}$, such that $||\overline{F}|| \leq F_{\mathrm{req}}$. Graphically, this set $\mathrm{NW_{req}}$ is simply a sphere centered at the origin of the $F_x - F_y - F_z$ frame with radius F_{req}. Fig. 6 illustrates an example manipulator [Fig. (6a)], its spherical required net wrench set $\mathrm{NW_{req}}$, and its parallelepiped available net wrench set $\mathrm{NW_{avail}}$ [Fig. (6b)] at that pose. It can be determined geometrically whether this pose is wrench-feasible by simply testing whether the distances between the facets of $\mathrm{NW_{avail}}$ and the origin are greater than or equal to F_{req}. In Fig. 6, $\mathrm{NW_{req}}$ is completely contained within $\mathrm{NW_{avail}}$; thus, this end-effector pose is wrench-feasible, and is therefore contained within the WFW of this manipulator.

This interpretation of wrench feasibility leads to an analytical method of forming the WFW by generating the boundaries of the workspace analytically. Consider a pose of a manipulator where $\mathrm{NW_{req}}$ is contained in $\mathrm{NW_{avail}}$ and is contacting one of the sides of $\mathrm{NW_{avail}}$. A small change in the pose of the manipulator can cause the pose to remain wrench-feasible (if $\mathrm{NW_{req}}$ remains inside $\mathrm{NW_{avail}}$) or to become not wrench-feasible (if part of $\mathrm{NW_{req}}$ is now outside of $\mathrm{NW_{avail}}$). Thus, this pose of the manipulator must be on the boundary of the WFW, because it is a point of transition between being wrench-feasible and no longer being wrench-feasible. Thus, the boundaries of the WFW consist of the set of all poses of the manipulator such that $\mathrm{NW_{req}} \subseteq \mathrm{NW_{avail}}$ and one or more of the planes bounding $\mathrm{NW_{avail}}$ contact $\mathrm{NW_{req}}$.

This situation can be represented as conditions on the geometry of the pose. Each facet of $\mathrm{NW_{avail}}$ can be expressed as a function of the cable wrenches $\$_1^w$ through $\$_p^w$, which are functions of the pose P. The condition of contact between one of these facets and $\mathrm{NW_{req}}$ results in a relationship between the wrenches $\$_1^w, \$_2^w, \ldots, \$_p^w$. This relationship is the condition for a pose being on the boundary of the workspace. Thus, these relationships represent an analytical definition of a boundary of the WFW. Repeating this process for each of the facets of $\mathrm{NW_{avail}}$ results in a set of analytical expressions that define the WFW boundaries.

Sections V and VI now formulate the boundaries of the WFW for two cases: planar and spatial cable robots and point-mass cable robots, respectively. Section V assumes a polyhedral $\mathrm{NW_{req}}$ for general cable robots. Section VI assumes a spherical $\mathrm{NW_{req}}$ for point-mass cable robots, as was the case for the example manipulator in Fig. 6.

V. WFW Analysis for Planar and Spatial Cable Robots

For planar and spatial cable robots, the wrench-exertion requirements may vary greatly from task to task. For example, one task may primarily require large moments to be exerted with very small associated forces, while another task may primarily require large forces to be exerted with very small associated moments. Some tasks may require large horizontal forces and small vertical forces, while others may require large vertical forces and small horizontal forces. Thus, it is not easy to choose a single geometry of $\mathrm{NW_{req}}$ that is representative of the various possible task requirements.

Therefore, in order to accommodate a wide variety of wrench-exertion requirements, $\mathrm{NW_{req}}$ will be assumed to be defined by an arbitrary polyhedron (or collection of polyhedra) with a finite number of vertices.[3] This allows a great deal of flexibility when specifying $\mathrm{NW_{req}}$, as nearly any arbitrary geometry of $\mathrm{NW_{req}}$ can be closely approximated by a collection of polyhedra.

Given such a geometry for $\mathrm{NW_{req}}$, the question now is how to test if a pose is wrench-feasible. It is shown in the following theorem that in order to test if $\mathrm{NW_{req}} \subseteq \mathrm{NW_{avail}}$, we only need to check that V, the set of vertices of $\mathrm{NW_{req}}$, is inside $\mathrm{NW_{avail}}$.

Theorem: If $\mathrm{NW_{req}}$ is a collection of a finite number of bounded polyhedra, each of which has a finite number of vertices, and if the set of vertices for the polyhedra is V, then

$$\mathrm{NW_{req}} \subseteq \mathrm{NW_{avail}} \quad \Leftrightarrow \quad V \subseteq \mathrm{NW_{avail}}. \tag{6}$$

Proof: Included in Appendix I.

Recall that the WFW boundaries are the set of all poses of the manipulator, such that $\mathrm{NW_{req}} \subseteq \mathrm{NW_{avail}}$ and one or more of the planes bounding $\mathrm{NW_{avail}}$ contact $\mathrm{NW_{req}}$. Because $\mathrm{NW_{avail}}$ is convex, if a plane bounding $\mathrm{NW_{avail}}$ contacts $\mathrm{NW_{req}}$, it must contact it at one or more vertices. **Thus, the set of boundary equations is simply the set of all expressions for contact between a vertex of** $\mathrm{NW_{req}}$ **and a facet of** $\mathrm{NW_{avail}}$.

Consider Fig. 7(a), which illustrates an example of a polyhedral $\mathrm{NW_{req}}$ (in this case, a cube) contained inside an $\mathrm{NW_{avail}}$. In order to form the WFW boundaries, it is necessary to form the set of equations that describe the condition of contact between a vertex of $\mathrm{NW_{req}}$ and a facet/side of $\mathrm{NW_{avail}}$.

The boundary equations can be formulated in determinant form. Consider the illustration in Fig. 7(b) of contact between a lower side of $\mathrm{NW_{avail}}$ and vertex v_1. The vertex contacts the

[3]It is assumed that none of the vertices of the polyhedra are located at infinity, and thus $\mathrm{NW_{req}}$ is actually assumed to be a set of *polytopes*, or bounded polyhedra [29].

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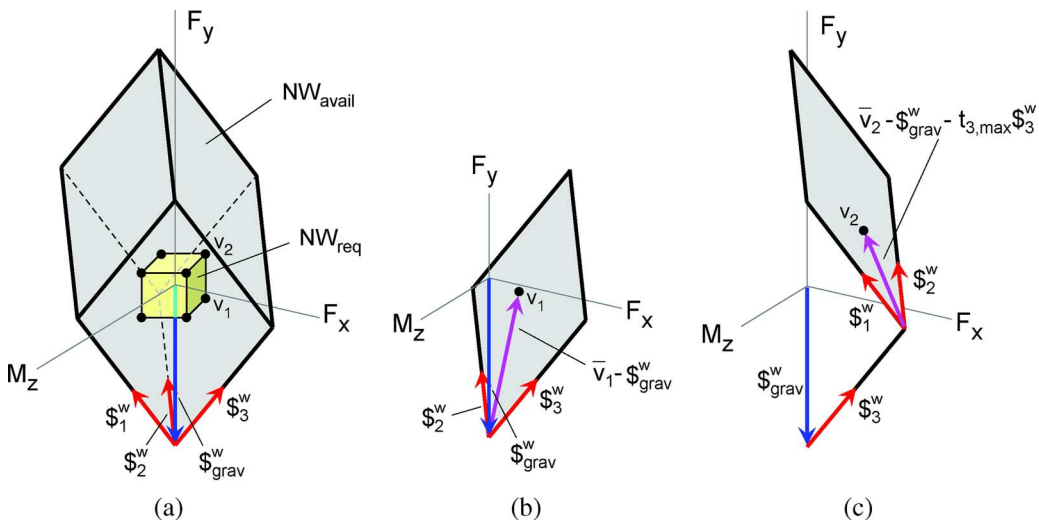

Fig. 7. Example available net wrench set containing its polyhedral required net wrench set. (a) Example polyhedral $\mathrm{NW_{req}}$ contained in $\mathrm{NW_{avail}}$. (b) Contact between a lower side of $\mathrm{NW_{avail}}$ and vertex v_1. (c) Contact between an upper side of $\mathrm{NW_{avail}}$ and vertex v_2.

lower side if the vector $(\overline{v}_1 - \$^w_{\mathrm{grav}})$ is a linear combination of $\$^w_2$ and $\$^w_3$, which can be expressed as

$$\det\left[\$^w_2 \quad \$^w_3 \quad \left(\overline{v}_1 - \$^w_{\mathrm{grav}}\right)\right] = 0. \tag{7}$$

Similarly, Fig. 7(c) illustrates contact between an upper side of $\mathrm{NW_{avail}}$ and vertex v_2. The vertex contacts the lower side if the vector $(\overline{v}_2 - \$^w_{\mathrm{grav}} - t_{3,\max}\$^w_3)$ is a linear combination of $\$^w_1$ and $\$^w_2$, which can be expressed as

$$\det\left[\$^w_1 \quad \$^w_2 \quad \left(\overline{v}_2 - \$^w_{\mathrm{grav}} - t_{3,\max}\$^w_3\right)\right] = 0. \tag{8}$$

In general, for an n-dimensional task space, a side S of $\mathrm{NW_{avail}}$ is spanned by $(n-1)$ wrenches. Let E be the set of wrenches along the edges of $\mathrm{NW_{avail}}$ that must be traced to get from the bottom of $\mathrm{NW_{avail}}$ (which is $\$^w_{\mathrm{grav}}$) to the bottom of side S, and let SP be the set of $(n-1)$ wrenches that positively span S. Then, S can be expressed as

$$S = \left\{ \$^w : \$^w = \sum_{\$^w_e \in E} t_{e,\max}\$^w_e + \sum_{\$^w_j \in SP} a_j t_{j,\max}\$^w_j, \quad 0 \le a_j \le 1 \right\}. \tag{9}$$

As an example, again consider Fig. 7. In the case of Fig. 7(b), $\$^w_2$ and $\$^w_3$ span the side we are interested in, and the bottom of the side is the same as the bottom of $\mathrm{NW_{avail}}$. Thus, $E = \{\emptyset\}$ and $SP = \{\$^w_2, \$^w_3\}$. In the case shown in Fig. 7(c), $\$^w_1$ and $\$^w_2$ span the side we are interested in, and in order to get from the bottom of $\mathrm{NW_{avail}}$ to the bottom of the side, we must traverse the edge that is along $\$^w_3$. Thus, $E = \{\$^w_3\}$ and $SP = \{\$^w_1, \$^w_2\}$.

Thus, the boundary equation corresponding to contact between any vertex $\overline{v}$ and a side of $\mathrm{NW_{avail}}$ is of the form

$$\det\left[\$^w_a \quad \$^w_b \quad \cdots \quad \$^w_i \left(\overline{v} - \$^w_{\mathrm{grav}} - \sum_{\$^w_j \in E} t_{j,\max}\$^w_j\right)\right] = 0 \tag{10}$$

where $\{\$^w_a, \$^w_b, \ldots, \$^w_i\} = SP$. In addition, we define the *lower boundaries* as the boundaries with $E = \{\emptyset\}$, and the *upper boundaries* as the boundaries with $E \neq \{\emptyset\}$.

Note that (10) must be formed for *every* vertex $\overline{v}$ of $\mathrm{NW_{req}}$ contacting *each* of the sides of $\mathrm{NW_{avail}}$. Thus, if there are w different vertices of $\mathrm{NW_{req}}$, then $2w\binom{p}{q-1}$ boundary equations must be formed. In the example shown in Fig. 7, $\mathrm{NW_{avail}}$ has 6 sides and $\mathrm{NW_{req}}$ has 8 vertices, thus 48 boundary equations must be formed. Clearly, the number of boundaries that must be formed is relatively large, even for simple geometries of $\mathrm{NW_{req}}$. Thus, if $\mathrm{NW_{req}}$ is complicated, a simplified approximation of $\mathrm{NW_{req}}$ with fewer vertices will make the computations more manageable.

However, the number of boundaries that must be formed can be reduced in some cases. For example, in Fig. 7(a), it is not possible for any of the lower sides of $\mathrm{NW_{avail}}$ to contact any of the upper vertices of $\mathrm{NW_{req}}$ without first contacting one of the lower vertices of $\mathrm{NW_{req}}$. Thus, boundary equations corresponding to contact with the upper vertices can be neglected. In addition, if the upper tension limits of the cables are high, the geometry of the WFW is dominated by the lower boundaries (i.e., those that correspond to a cable having zero tension). Thus, in some cases, it may only be necessary to form a few of the workspace boundaries in order to determine the majority of the geometry of the workspace. Further research is needed on a systematic way to determine the minimum necessary set of boundary equations.

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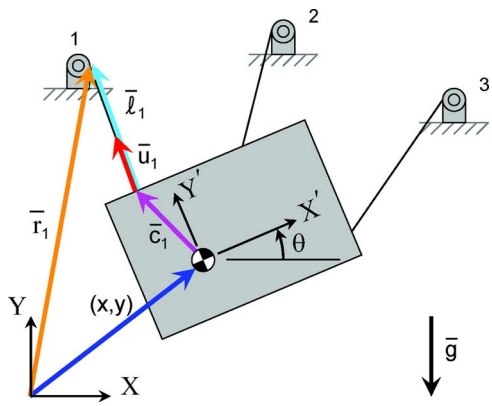

Fig. 8. Kinematic parameters for a planar cable robot.

A. WFW of Planar Cable Robots

1) Forming Lower WFW Boundaries: Consider a general planar cable robot where the wrench-exertion requirements of a task are defined by a polyhedral $\mathrm{NW}_{\mathrm{req}}$. Let the set of all vertices of $\mathrm{NW}_{\mathrm{req}}$ be V. As shown in Fig. 8, the pose of the end-effector is defined as (x, y, θ), where (x, y) is the position of the center of gravity of the end-effector in the fixed global coordinate frame $X - Y$, and θ is the rotation of the end-effector, defined as the relative angle between the moving coordinate frame $X' - Y'$ attached to the end-effector and the global coordinate frame $X - Y$. Without loss of generality, the fixed coordinate frame can be chosen such that the Y axis is vertical (aligned with gravity), the X axis points to the right, and counterclockwise rotations (and moments) are considered positive.

The location of motor i (or location of the pulley through which the cable is routed) with respect to the fixed global frame is $\bar{r}_i$, and the vectors from (x, y) to the attachment point of the cable i is $\bar{c}_i$. The notation used for these vectors is as follows:

$$\bar{r}_i = \begin{pmatrix} r_{i,x} \\ r_{i,y} \end{pmatrix} \quad \bar{c}_i = \begin{pmatrix} c_{i,x} \\ c_{i,y} \end{pmatrix}. \tag{11}$$

Note that the vector $\bar{r}_i$ is defined in the fixed global coordinate frame, while $\bar{c}_i$ is defined in the moving coordinate frame attached to the end-effector. Thus, both vectors are constant vectors (i.e., they do not change in their respective frames as the end-effector moves).

Let us define the cable length vector $\bar{\ell}_i$ in the global coordinate frame as the vector along cable i directed from the cable attachment points to the corresponding motor locations

$$\bar{\ell}_i = \bar{r}_i - \left[\begin{pmatrix} x \\ y \end{pmatrix} + \boldsymbol{R}(\theta)\bar{c}_i \right] \tag{12}$$

where

$$\boldsymbol{R}(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}. \tag{13}$$

Note that the length of cable i is $\|\bar{\ell}_i\|$.

The unit vector $\bar{u}_i$ along cable i directed away from the end-effector is then defined as

$$\bar{u}_i = \frac{\bar{\ell}_i}{\|\bar{\ell}_i\|}. \tag{14}$$

When $\|\bar{\ell}_i\| = 0$, the corresponding unit vector is undefined, and is chosen to be $\bar{0}$.

Using (10), the equation for any one of the lower boundaries is of the form

$$\det\left[\$_i^w \quad \$_j^w \quad \left(\bar{v} - \$_{\mathrm{grav}}^w\right)\right] = 0. \tag{15}$$

Let the wrench corresponding to vertex $\bar{v}$ be expressed as $\bar{v} = (F_x, F_y, M)^T$. Then using (3) and noting that here $\$_{\mathrm{grav}}^w = (0, -mg, 0)^T$, (15) becomes[4]

$$\det\begin{bmatrix} u_{i,x} & u_{j,x} & F_x \\ u_{i,y} & u_{j,y} & F_y + mg \\ (\boldsymbol{R}(\theta)\bar{c}_i) \times \bar{u}_i & (\boldsymbol{R}(\theta)\bar{c}_j) \times \bar{u}_j & M \end{bmatrix} = 0. \tag{16}$$

Evaluation of (16) results in the following:

$$\begin{aligned} & u_{i,x}u_{j,y}M + u_{jx}(F_y + mg)\left((\boldsymbol{R}(\theta)\bar{c}_i) \times \bar{u}_i\right) + \cdots \\ & \quad + F_x u_{i,y}\left((\boldsymbol{R}(\theta)\bar{c}_j) \times \bar{u}_j\right) \\ & \quad - u_{i,x}(F_y + mg)\left((\boldsymbol{R}(\theta)\bar{c}_j) \times \bar{u}_j\right) - \cdots \\ & \quad - u_{j,x}u_{i,y}M - F_x u_{j,y}\left((\boldsymbol{R}(\theta)\bar{c}_i) \times \bar{u}_i\right) = 0. \end{aligned} \tag{17}$$

Substituting for $\bar{u}_i$ and $\bar{u}_j$ in terms of $\bar{\ell}_i$ and $\bar{\ell}_j$, as defined in (14), results in each term having a common factor of $1/\|\bar{\ell}_i\| \cdot \|\bar{\ell}_j\|$. Thus, multiplying both sides of (17) by $\|\bar{\ell}_i\| \cdot \|\bar{\ell}_j\|$ gives

$$\begin{aligned} & \ell_{i,x}\ell_{j,y}M + \ell_{jx}(F_y + mg)\left((\boldsymbol{R}(\theta)\bar{c}_i) \times \bar{\ell}_i\right) + \cdots \\ & \quad + F_x \ell_{i,y}\left((\boldsymbol{R}(\theta)\bar{c}_j) \times \bar{\ell}_j\right) \\ & \quad - \ell_{i,x}(F_y + mg)\left((\boldsymbol{R}(\theta)\bar{c}_j) \times \bar{\ell}_j\right) - \cdots \\ & \quad - \ell_{j,x}\ell_{i,y}M - F_x \ell_{j,y}\left((\boldsymbol{R}(\theta)\bar{c}_i) \times \bar{\ell}_i\right) = 0 \end{aligned} \tag{18}$$

which eliminates the complexity of the square-root and quadratic terms in $\|\bar{\ell}_i\|$ and $\|\bar{\ell}_j\|$. Each of the terms in (18) can now be expressed in terms of known constants ($\bar{r}_i$, $\bar{r}_j$, $\bar{c}_i$, $\bar{c}_j$, mg, F_x, F_y, and M) and the variables of interest (x, y, and θ). Collecting terms results in the equation for the lower boundaries of the WFW for planar cable robots

$$\boxed{a_1x^2 + a_2x + a_3xy + a_4y + a_5y^2 + a_6 = 0} \tag{19}$$

where each a_i is a function of θ and known constants. These coefficients are included in Appendix II.

Because these coefficients are fairly complicated, it is not trivial to plot this workspace boundary. However, if the manipulator is considered at a known constant orientation, each a_i becomes a constant, and the boundary equation reduces to a relatively simple polynomial in x and y. Thus, it is possible

[4]While the cross-product operation is only strictly defined for 3-D vectors, it is used here on 2-D vectors in order to retain the same form of the boundary equations as that obtained for the spatial case. In two dimensions, the operation $\bar{a} \times \bar{b}$ is defined as $\bar{a} \times \bar{b} = \det[\bar{a}\ \bar{b}]$.

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to easily visualize the workspace boundaries by plotting multiple constant orientation boundary curves, essentially viewing the workspace boundary one "slice" at a time.

Special Case: If instead we would like to generate the SEW ($\mathrm{NW}_{\mathrm{req}} = (F_x, F_y, M)^T = (0,0,0)^T$), the coefficients of (19) simplify such that the workspace boundary equation can be put in the form $y = g(x,\theta)$, where

$$y = g(x,\theta) = -\frac{a_1x^2 + a_2x + a_6}{a_3x + a_4}. \tag{20}$$

Again, if we consider a constant orientation of the end-effector, each a_i becomes a constant and the constant-orientation boundary has a polynomial form. Thus, the lower boundaries of the SEW can be plotted easily for constant orientation of a planar cable robot.

2) Forming Upper WFW Boundaries: Using (10), an upper boundary of a planar cable robot has the form

$$\det\left[\$_i^w \quad \$_j^w \quad \left(\overline{v} - \$_{\mathrm{grav}}^w - \sum_{\$_k^w \in E} t_{k,\max}\$_k^w\right)\right] = 0. \tag{21}$$

Let $E = \{\$_\alpha^w, \$_\beta^w, \ldots\}$. Then using the previously defined notation, the boundary equation becomes

$$\det\begin{bmatrix} u_{i,x} & u_{j,x} & b_1 \\ u_{i,y} & u_{j,y} & b_2 \\ (\boldsymbol{R}(\theta)\overline{c}_i)\times\overline{u}_i & (\boldsymbol{R}(\theta)\overline{c}_j)\times\overline{u}_j & b_3 \end{bmatrix} = 0 \tag{22}$$

where

$$\begin{aligned} b_1 &= F_x + t_{\alpha,\max}u_{\alpha,x} + t_{\beta,\max}u_{\beta,x} + \cdots \\ b_2 &= F_y + mg + t_{\alpha,\max}u_{\alpha,y} + t_{\beta,\max}u_{\beta,y} + \cdots \\ b_3 &= M + t_{\alpha,\max}(\boldsymbol{R}(\theta)\overline{c}_\alpha)\times\overline{u}_\alpha + t_{\beta,\max}(\boldsymbol{R}(\theta)\overline{c}_\beta)\times\overline{u}_\beta + \cdots \end{aligned}$$

Evaluation of (22) results in an expression similar in form to (17), and substitutions can be made for $\overline{u}_i$, $\overline{u}_j$, $\overline{u}_\alpha$, $\overline{u}_\beta$, etc., in terms of $\overline{\ell}_i$, $\overline{\ell}_j$, $\overline{\ell}_\alpha$, $\overline{\ell}_\beta$, etc. However, unlike (17), the factor of $1/\|\overline{\ell}_i\| \cdot \|\overline{\ell}_j\| \cdot \|\overline{\ell}_\alpha\| \cdot \|\overline{\ell}_\beta\| \cdot \ldots$ is not common to all terms, and therefore, cannot be canceled. Thus, the resulting boundary equation will no longer have polynomial form, and will include many square-root terms.

Because of this amount of complexity, it does not appear useful to fully detail the analytical form of the upper WFW boundaries, as the resulting equations will not provide much insight into the geometry of the boundaries and will likely need to be plotted using numerical techniques. However, for both underconstrained and fully constrained manipulators, if the upper tension limits are relatively high, then the lower WFW boundaries determine the majority of the geometry of the WFW. Thus, the lower workspace boundaries found previously are the key boundaries to consider.

B. Example Planar WFW

As an example, let us examine the WFW of the manipulator shown in Fig. 9. The pose of the manipulator is $(x, y, \theta)^T$, where

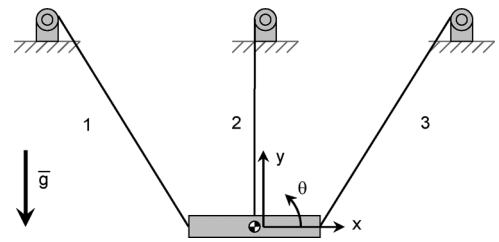

Fig. 9. Example manipulator (note: not drawn to scale).

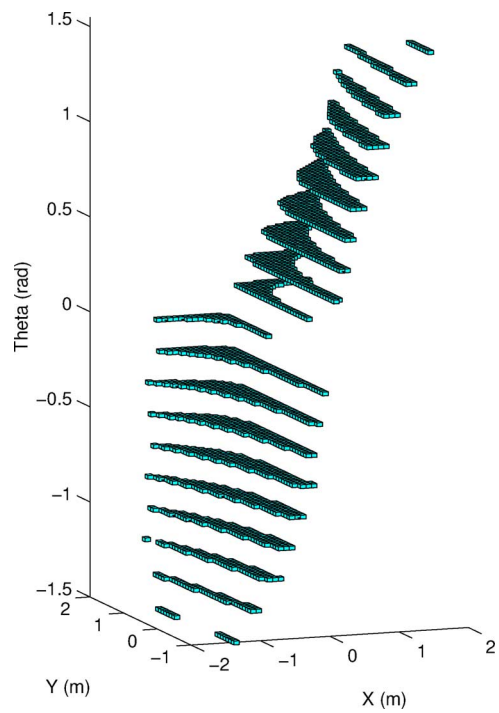

Fig. 10. Numerically determined SEW for an example manipulator.

the pose of the manipulator as shown in Fig. 9 is (0 m, 0 m, 0 rad)^T. Motor 1 is located at $(-1.5, 2)^T$ m, motor 2 is located at $(0, 2)^T$ m, and motor 3 is located at $(1.5, 2)^T$ m. The vectors from the center of gravity to the cable attachment points are $\overline{c}_1 = (-0.3, 0)^T$ m, $\overline{c}_2 = (0, 0.05)^T$ m, and $\overline{c}_3 = (0.3, 0)^T$ m. The weight of the end-effector is 20 N, and the cables are assumed to have very high upper tension limits. We will now generate the WFW numerically for two choices of $\mathrm{NW}_{\mathrm{req}}$, and compare the results with the analytically determined WFW boundaries.

The WFW is first calculated for $\mathrm{NW}_{\mathrm{req}} = (0, 0, 0(1/\mathrm{m}))^T$ N (i.e., the SEW) numerically using MATLAB, where the taskspace is discretized and searched exhaustively. The resulting discretized workspace is shown in Fig. 10. Note that the workspace is continuous in the angle θ, but is discretized into slices here to better show the interior shape of the workspace.

Now let us compare the numerical results with analytical results for the workspace boundaries. In this case, there is only one vertex to consider. Because the upper tension limits of the cables are very high, only the lower workspace boundaries will be considered. For simplicity, we will only consider a constant

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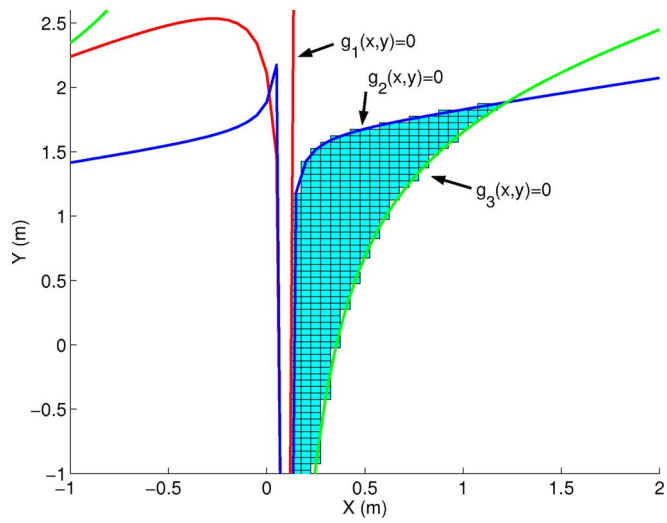

Fig. 11. Numerically determined WFW and analytically determined WFW boundaries for an example manipulator at a constant orientation of $\theta = \pi/8$ and $\mathrm{NW_{req}} = \{(0,0,0(1/\mathrm{m}))^T\ \mathrm{N}\}$.

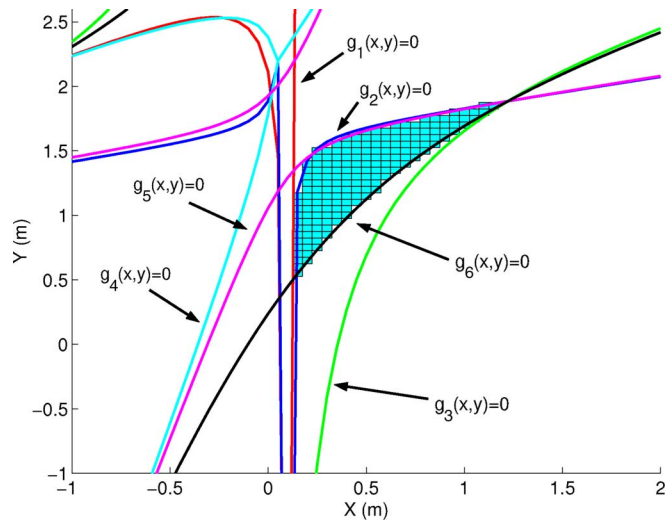

Fig. 12. Numerically determined WFW and analytically determined WFW boundaries for an example manipulator at a constant orientation of $\theta = \pi/8$ and $\mathrm{NW_{req}} = conv\{(0,0,0(1/\mathrm{m}))^T\ \mathrm{N}, (5,0,0(1/\mathrm{m}))^T\ \mathrm{N}\}$.

orientation "slice" of the workspace at $\theta = \pi/8$. Using (19) and (32)–(37), the three boundary equations are

$$\begin{aligned} g_1(x,y) &= 3.22x^2 + 14.851x - 5.161xy + 0.574y - 1.214 = 0 \\ g_2(x,y) &= 1.372x^2 + 9.494x - 5.926xy + 0.574y - 1.082 = 0 \\ g_3(x,y) &= 4.592x^2 + 22.173x - 11.087xy - 8.423 = 0 \end{aligned}$$

where $g_1(x,y) = 0$ is the boundary equation corresponding to $SP = \{\$_1^w, \$_2^w\}$, $g_2(x,y) = 0$ is the boundary equation corresponding to $SP = \{\$_2^w, \$_3^w\}$, and $g_3(x,y) = 0$ is the boundary equation corresponding to $SP = \{\$_1^w, \$_3^w\}$. The boundary equations are plotted with the discretized workspace in Fig. 11. Note that the analytically determined boundaries agree exactly with the bounds of the numerically determined workspace.

Now let us change $\mathrm{NW_{req}}$ to a polyhedral with two vertices (i.e., a line segment). Let the two vertices be the origin and a pure force of 5 N to the right. Then $\mathrm{NW_{req}} = \mathrm{conv}\{(0,0,0(1/\mathrm{m}))^T\ \mathrm{N}, (5,0,0(1/\mathrm{m}))^T\ \mathrm{N}\}$. In this case, there are two vertices to consider, thus we must form six boundary equations, three for each vertex. Because one of the vertices is the origin, which was the vertex used in the previous case, the first three equations are $g_1(x,y) = 0$, $g_2(x,y) = 0$, and $g_3(x,y) = 0$, as given previously. Using (19) and (32)–(37), the three additional boundary equations are

$$\begin{aligned} g_4(x,y) &= 3.22x^2 + 16.461x - 5.966xy \\ &\quad - 5.373y + 1.29y^2 + 5.479 = 0 \\ g_5(x,y) &= 1.372x^2 + 10.18x - 6.269xy \\ &\quad - 4.416y + 1.481y^2 + 3.012 = 0 \\ g_6(x,y) &= 4.592x^2 + 24.469x - 12.235xy \\ &\quad - 11.087y + 2.772y^2 + 2.466 = 0 \end{aligned}$$

where $g_4(x,y) = 0$ is the boundary equation corresponding to $SP = \{\$_1^w, \$_2^w\}$, $g_5(x,y) = 0$ is the boundary equation corresponding to $SP = \{\$_2^w, \$_3^w\}$, and $g_6(x,y) = 0$ is the boundary equation corresponding to $SP = \{\$_1^w, \$_3^w\}$. The boundary equations are plotted with the discretized workspace in Fig. 12. Note that the analytically determined boundaries again agree exactly with the bounds of the numerically determined workspace.

C. WFW of Spatial Cable Robots

The formulation of the WFW boundaries for spatial cable robots is quite similar to that of planar cable robots. Using (10), each boundary equation is of the form

$$\det\left[\$_a^w\ \$_b^w\ \$_c^w\ \$_d^w\ \$_e^w\ \left(\overline{v} - \$_{\mathrm{grav}}^w - \sum_{\$_j^w \in E} t_{j,\max}\$_j^w\right)\right] = 0 \tag{23}$$

where $\{\$_a^w, \$_b^w, \ldots, \$_e^w\} = SP$. Note that because the manipulator is spatial, this is a 6×6 matrix (i.e., SP always contains five wrenches). Also, note that each wrench $\$^w$ is a function of both the position of the end-effector (x,y,z) and of the orientation of the end-effector, expressed here in Euler angles (ψ,θ,ϕ).

It would be desirable to expand (23) in terms of known system parameters and the pose of the manipulator (x,y,z,ψ,θ,ϕ), similar to what was done for the planar case. However, even in the simpler case of forming lower boundaries, evaluating (23) results in a fifth-order polynomial equation in x, y, and z, where each of the 56 polynomial coefficients is a function of ψ, θ, and ϕ

$$\begin{aligned} &a_1x^5 + a_2x^4y + a_3x^4z + a_4x^4 + \cdots \\ &\quad \cdots + a_{52}xz^4 + a_{53}yz^4 + a_{54}z^4 + a_{55}z^5 + a_{56} = 0. \end{aligned} \tag{24}$$

Thus, even for the simple case of the lower workspace boundaries, the resulting polynomial expressions are very complicated, and thus are not detailed here. However, if desired, it would be possible to use a symbolic manipulation program to perform the calculation of the coefficients.

As was the case for planar cable robots, the upper WFW boundaries for spatial cable robots are more complicated, due

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to the inclusion of $||\ell_i||$ terms. As a result, it is unlikely that forming these boundaries analytically will be useful. Again, if the upper tension limits of the cables are very high, the geometry of the WFW will be largely determined by the lower workspace boundaries.

VI. WFW Analysis for Point-Mass Cable Robots

Point-mass cable robots can only exert forces on their surroundings (and no moments), thus for point-mass cable robots, $\mathrm{NW}_{\mathrm{req}}$ is a set of pure forces. The most common scenario is that the manipulator needs to be able to exert a required force F_{req} in any direction. Graphically, this set $\mathrm{NW}_{\mathrm{req}}$ is simply a sphere centered at the origin with radius F_{req}, as was shown in Fig. 6. While the workspace analysis in Section V applies to point-mass cable robots, a polyhedral approximation of a sphere will require many workspace boundaries to be calculated. Thus, the special case of a point-mass cable robot with a spherical $\mathrm{NW}_{\mathrm{req}}$ is considered here.

A. Forming the WFW Boundaries

Given a 3-D point-mass cable robot with three cables ($p = 3$)[5] in a particular end-effector pose, the available net wrench set is known to be a parallelepiped described by $\mathrm{NW}_{\mathrm{avail}} = \{\$^w : \$^w = a_1 t_{1,\max}\overline{u}_1 + a_2 t_{2,\max}\overline{u}_2 + a_3 t_{3,\max}\overline{u}_3 + m\overline{g},\ 0 \le a_i \le 1\}$, as shown in Fig. 6(b). The required net wrench set $\mathrm{NW}_{\mathrm{req}}$ is assumed to be a sphere, with radius F_{req} centered at the origin.

Based on the conclusions made earlier, at every pose on the boundary of the WFW, at least one side of $\mathrm{NW}_{\mathrm{avail}}$ contacts $\mathrm{NW}_{\mathrm{req}}$. In this case, that means a side of $\mathrm{NW}_{\mathrm{avail}}$ is tangent to the spherical $\mathrm{NW}_{\mathrm{req}}$. A set of six vectors can be defined, where each vector is the shortest vector from the origin to one of the six sides of $\mathrm{NW}_{\mathrm{avail}}$ (orthogonal distance vectors), as illustrated in Fig. 13.[6] For each of the lower three sides of $\mathrm{NW}_{\mathrm{avail}}$, the vector $\overline{d}_{ij}$ is directed towards the lower side spanned by $\overline{u}_i$ and $\overline{u}_j$. For each of the three upper sides of $\mathrm{NW}_{\mathrm{avail}}$, the vector $\overline{d}'_{ij}$ is directed towards the upper side spanned by $\overline{u}_i$ and $\overline{u}_j$.

Because $\mathrm{NW}_{\mathrm{req}}$ is a sphere of radius F_{req}, an intersection with the boundary of $\mathrm{NW}_{\mathrm{avail}}$ will occur whenever $||\overline{d}_{ij}|| \le F_{\mathrm{req}}$ or $||\overline{d}'_{ij}|| \le F_{\mathrm{req}}$. Thus, an end-effector position is in the WFW if and only if:

$$||\overline{d}_{ij}|| \ge F_{\mathrm{req}} \quad \text{and} \quad \left\|\overline{d}'_{ij}\right\| \ge F_{\mathrm{req}}, \quad \text{for } \{i,j\} \in \{\{1,2\},\{2,3\},\{3,1\}\}. \tag{25}$$

[5]Note that this procedure can be extended to a point-mass cable robot with any number of cables.

[6]For ease of visualization, this is illustrated for the 2-D case, where four orthogonal distance vectors $(\overline{d}_1, \overline{d}_2, \overline{d}'_1, \overline{d}'_2)$ are directed towards each of the four sides of a planar $\mathrm{NW}_{\mathrm{avail}}$.

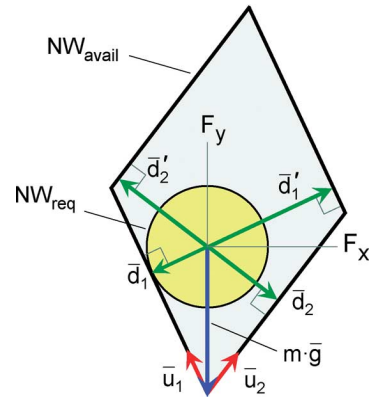

Fig. 13. Available net wrench set with orthogonal distance vectors.

By forming a series of vector loop equations involving known vectors, the magnitudes $||\overline{d}_{ij}||$ and $||\overline{d}'_{ij}||$ can be calculated. For brevity, the details of the derivation are not included here, but can be found in [30]. If $\overline{u}_i = (u_{ix}, u_{iy}, u_{iz})^T$, motor mount location i is $\overline{r}_i = (r_{ix}, r_{iy}, r_{iz})^T$, and the end-effector location is $\overline{ee} = (x, y, z)^T$, then solving the vector loop equations for each d_{ij} and d'_{ij} results in (26) and (27), shown at the bottom of the page, where

$$u_{\ell x} = \frac{r_{\ell x} - x}{\sqrt{(r_{\ell x} - x)^2 + (r_{\ell y} - y)^2 + (r_{\ell z} - z)^2}} \tag{28}$$

$$u_{\ell y} = \frac{r_{\ell y} - y}{\sqrt{(r_{\ell x} - x)^2 + (r_{\ell y} - y)^2 + (r_{\ell z} - z)^2}} \tag{29}$$

$$u_{\ell z} = \frac{r_{\ell z} - y}{\sqrt{(r_{\ell x} - x)^2 + (r_{\ell y} - z)^2 + (r_{\ell z} - z)^2}} \quad \text{for } \ell = 1,2,3. \tag{30}$$

Note that $\cos^{-1}$ is assumed to yield a result between zero and π.

The boundaries of the WFW can now be expressed analytically by substituting identities (28)–(30) into (26) and (27), and setting each distance $||\overline{d}_{ij}||$ and $||\overline{d}'_{ij}||$ equal to F_{req}. Equations (26) and (27) represent six implicit expressions of the six boundaries of the WFW.

B. Analysis of Boundary Equations

From the equations for these WFW boundaries, several workspace properties and trends can be observed. These will be listed briefly here, but full derivations of the properties and workspace trends are included in [30] and [31]. In addition, details of the workspace derivation for a two-cable planar point-mass cable robot are also included in [30] and [31].

1) Workspace Properties: Based on the expressions for the boundaries of the WFW, the following properties can be observed for the 3-D case. The properties of the 2-D case (a planar

$$||\overline{d}_{ij}|| = \frac{mg(u_{jx}u_{iy} - u_{ix}u_{jy})}{\sqrt{(u_{iy}u_{jz} - u_{jy}u_{iz})^2 + (u_{jx}u_{iz} - u_{ix}u_{jz})^2 + (u_{ix}u_{jy} - u_{jx}u_{iy})^2}} \tag{26}$$

$$\left\|\overline{d}'_{ij}\right\| = t_{k,\max}\sin\left[\cos^{-1}(\overline{u}_j \cdot \overline{u}_k)\right]\sin\left[\cos^{-1}(\overline{u}_k \cdot \overline{u}_i)\right] - d_{ij} \quad \text{for } \{i,j,k\} \in \{\{1,2,3\},\{2,3,1\},\{3,1,2\}\} \tag{27}$$

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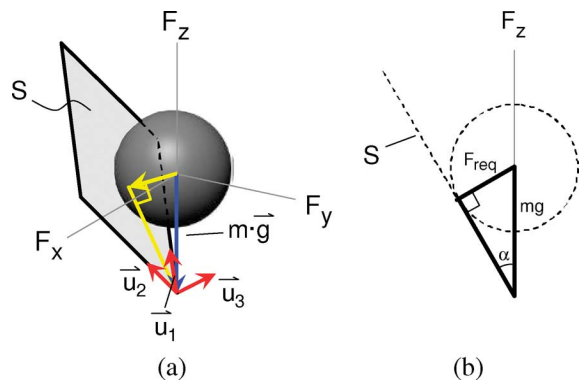

Fig. 14. Lower side S of $\mathrm{NW_{avail}}$ tangent to the spherical $\mathrm{NW_{req}}$, resulting in $\alpha = sin^{-1}(F_{\mathrm{req}}/mg)$. (a) Side S tangent to $\mathrm{NW_{req}}$. (b) Rotated view.

point-mass cable robot with circular $\mathrm{NW_{req}}$) are included in parentheses.

Property 1: Lower WFW boundaries, i.e., those defined by (26), are always planes. (For the 2-D case, the lower boundaries are lines.)

Proof: For end-effector positions on a lower boundary of the WFW, a lower side S of $\mathrm{NW_{avail}}$ is tangent to the spherical $\mathrm{NW_{req}}$, as shown in Fig. 14, where here S is spanned by $\overline{u}_1$ and $\overline{u}_2$. The plane containing S in the wrench space must therefore form an angle from vertical of $\alpha = \sin^{-1}(F_{\mathrm{req}}/mg)$ resulting from the right triangle formed in Fig. 14.

Given this geometric condition in the wrench space, the structure of the workspace boundary can be formed in the task space. Because $\overline{u}_i$ and $\overline{u}_j$ define the cable directions, the plane they span in the task space must pass through the motor mount locations *and* form an angle of α with vertical. These conditions are only satisfied for end-effector locations within this plane, thus each lower workspace boundary is a plane.□

Property 2: All lower workspace boundaries have the same relative angle from vertical. (For the 2-D case, the lower boundaries are lines with the same relative angle from vertical.)

Proof: As shown in the previous proof, each lower boundary of the workspace is a plane that forms an angle with vertical of $\alpha = \sin^{-1}(F_{\mathrm{req}}/mg)$.□

Property 3: Each workspace boundary must pass through exactly two motor mount locations.

Proof: Not included here due to space limitations. See [30] and [31] for proof.

These properties can be observed in Fig. 15(a)–15(c), which show the analytically computed WFW for 2-D manipulators with the elevation of motor 2 varied. Here, $\mathrm{NW_{req}}$ is a circle. In each figure, the four curves are the four workspace boundaries found using the 2-D versions of (26) and (27), and the shaded region is the resulting WFW. The lower two boundaries (the straight lines) come from the 2-D version of (26), and the upper two boundaries (the curved lines) come from the 2-D version of (27).

2) Workspace Geometry Trends: Given the analytical expressions for the workspace boundaries, it is possible to vary different design parameters and see how they affect the geometry of the WFW. When designing a point-mass cable robot, these trends can be used to adjust the manipulator design appropriately to achieve the desired workspace geometry. The specific trends are not included here, but [31] and [30] examine the effects of: 1) varying maximum cable tensions; 2) varying end-effector mass; 3) varying the radius of $\mathrm{NW_{req}}$; and 4) varying motor mount locations. Fig. 15(a)–15(c) illustrate how the geometry of the WFW changes as the elevation of one of the motors is varied. Note that in these plots, the upper tension limits $t_{\max,i}$ are relatively low, and as a result, the upper workspace boundaries have an increased impact on the workspace geometry. If $t_{\max,i} \gg mg$, the upper boundaries will approach a straight line from motor 1 to motor 2.

VII. Constructing Other Workspaces

One of the additional benefits of the method developed here for generating the WFW is that several previously proposed workspaces can be described using this theoretical framework. The SEW is actually a special case of the WFW where $\mathrm{NW_{req}} = \{\overline{0}\}$. The controllable workspace, defined in [14], is a special case of the WFW where $\mathrm{NW_{req}}$ is a single point in the wrench space. The dynamic workspace, defined in [25], is a special case of the WFW where a specific acceleration of the end-effector must be achieved, corresponding to a specific single wrench that must be exerted on the end-effector. Thus, the method presented in the previous sections for analytically forming the WFW can be used to analytically form each of these workspaces.

VIII. Summary and Conclusions

For manipulator tasks that require the end-effector to exert a specific set of wrenches, the WFW represents the usable workspace of the robot. The required net wrench set $\mathrm{NW_{req}}$ was defined to be the set of wrenches that the manipulator must exert, and the available net wrench set $\mathrm{NW_{avail}}$ was defined to be the set of wrenches that the manipulator could exert at a given pose.

Analytical expressions for the WFW boundaries for point-mass, planar, and spatial cable robots were then formed, assuming $\mathrm{NW_{req}}$ to be a collection of polyhedra for the planar and spatial case, and a sphere for the point-mass case. This analytical formulation of the WFW boundaries is a significant improvement over existing methods, which largely rely on numerical exhaustive-search approaches. The workspace boundary equations enable analysis of workspace properties and trends, as was performed in the point-mass case. This analysis can then be used to develop design guidelines for optimizing the geometry of the WFW for cable robots.

IX. Future Work

There are several topics for future work that can be done in this area. Because of the complexity of forming the upper WFW boundaries for planar and spatial cable robots, it is not currently feasible to formulate all workspace boundaries analytically. Thus, it would be advantageous to develop a more effective method for formulating the upper workspace boundaries. For example, an efficient numerical method could be developed for approximating the upper boundaries, which could be coupled with the analytically determined lower boundaries to form a more complete representation of the WFW boundaries.

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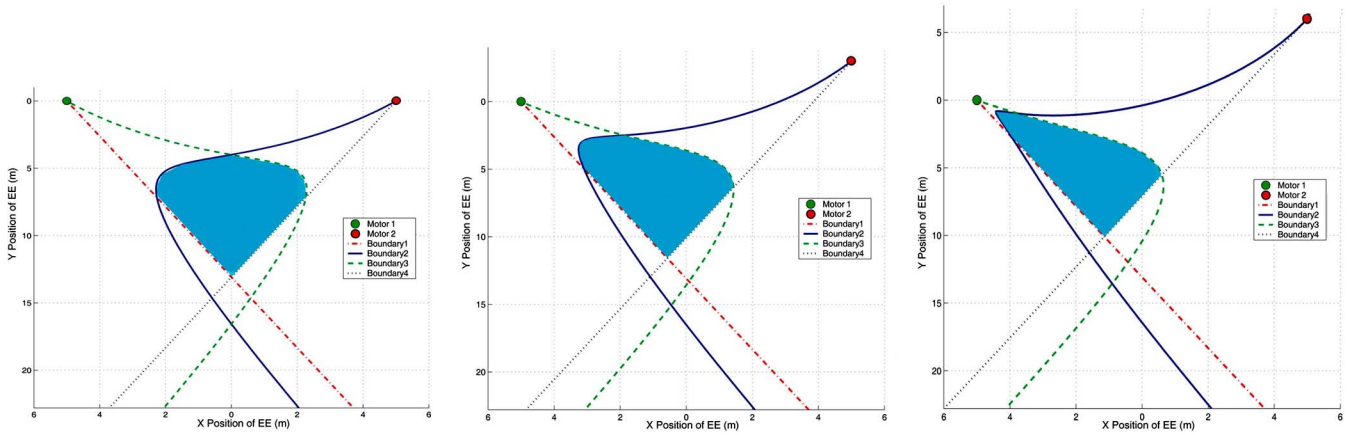

(a) (b) (c)

Fig. 15. Analytically computed WFW for planar two-cable robot with constant parameters $\overline{r}_1 = [-5, 0]$ m, $F_{\text{req}} = 7.1$ N, $t_{1,\max} = t_{2,\max} = 22.9$ N, $m = 2.0$ kg, and varied parameters (a) $\overline{r}_2 = [5, 0]$ m, (b) $\overline{r}_2 = [5, 3]$ m, and (c) $\overline{r}_2 = [5, 6]$ m.

In addition, it may be necessary to incorporate the effects of cable interference. Interference due to cables contacting each other and cables contacting the end-effector reduces the effective workspace. Thus, if analytical expressions were formulated for the condition of interference, these would constitute additional workspace boundaries.

Finally, it is planned to analyze the workspace properties and geometry trends of planar and spatial cable robots based on the results presented here. As was mentioned previously, this would allow design guidelines to be synthesized for designing a cable robot with a desired WFW geometry.

Appendix I
Proof of Theorem

Theorem: If NW_{req} is a collection of a finite number of bounded polyhedra,[7] each of which has a finite number of vertices, and if the set of vertices for the polyhedra is V, then

$$\text{NW}_{\text{req}} \subseteq \text{NW}_{\text{avail}} \quad \Leftrightarrow \quad V \subseteq \text{NW}_{\text{avail}}. \tag{31}$$

Proof: This proof contains two directions, the first of which is trivial.

Proof of $\text{NW}_{\text{req}} \subseteq \text{NW}_{\text{avail}} \Rightarrow V \subseteq \text{NW}_{\text{avail}}$:
$V \subseteq \text{NW}_{\text{req}}$, so from $\text{NW}_{\text{req}} \subseteq \text{NW}_{\text{avail}}$ it follows that $V \subseteq \text{NW}_{\text{avail}}$.

Proof of $\text{NW}_{\text{req}} \subseteq \text{NW}_{\text{avail}} \Leftarrow V \subseteq \text{NW}_{\text{avail}}$:
First, we must prove that NW_{avail} is convex. A set S is said to be convex if $\mu x + (1-\mu)y \in S$ for all $x, y \in S$ and $0 \le \mu \le 1$ [32]. Recall that $\text{NW}_{\text{avail}} = \{\$^w : \$^w = a_1 t_{1,\max}\$_1^w + \ldots + a_p t_{p,\max}\$_p^w + \$_{\text{grav}}^w,\ 0 \le a_i \le 1\}$, where $t_{i,\max}$ is the maximum allowable tension in cable i and $\$_i^w$ is the ith column of $\boldsymbol{J}^T$. Then, given two wrenches $\$_\alpha^w, \$_\beta^w \in \text{NW}_{\text{avail}}$

$$\begin{aligned}\$_\alpha^w &= a_{\alpha,1} t_{1,\max}\$_1^w + \cdots + a_{\alpha,p} t_{p,\max}\$_p^w + \$_{\text{grav}}^w \\ &\quad 0 \le a_{\alpha,i} \le 1 \\ \$_\beta^w &= a_{\beta,1} t_{1,\max}\$_1^w + \cdots + a_{\beta,p} t_{p,\max}\$_p^w + \$_{\text{grav}}^w \\ &\quad 0 \le a_{\beta,i} \le 1.\end{aligned}$$

Let $0 \le \mu \le 1$. Then let $\$_\gamma^w$ be a convex combination of $\$_\alpha^w$ and $\$_\beta^w$

$$\begin{aligned}\$_\gamma^w &= \mu\$_\alpha^w + (1-\mu)\$_\beta^w \\ &= \mu a_{\alpha,1} t_{1,\max}\$_1^w + \cdots + \mu a_{\alpha,p} t_{p,\max}\$_p^w + \mu\$_{\text{grav}}^w + \cdots \\ &\quad + (1-\mu) a_{\beta,1} t_{1,\max}\$_1^w + \cdots \\ &\quad + (1-\mu) a_{\beta,p} t_{p,\max}\$_p^w + (1-\mu)\$_{\text{grav}}^w \\ &= (\mu a_{\alpha,1} + (1-\mu) a_{\beta,1})\, t_{1,\max}\$_1^w + \cdots \\ &\quad + (\mu a_{\alpha,p} + (1-\mu) a_{\beta,p})\, t_{p,\max}\$_p^w + \$_{\text{grav}}^w.\end{aligned}$$

Because $0 \le a_{\alpha,i} \le 1$ and $0 \le a_{\beta,i} \le 1$

$$0 \le (\mu a_{\alpha,i} + (1-\mu) a_{\beta,i}) \le 1.$$

If we define a new set of coefficients $a_{\gamma,i}$ as

$$a_{\gamma,i} = (\mu a_{\alpha,i} + (1-\mu) a_{\beta,i}) \quad i = 1, \ldots, p$$

then

$$\$_\gamma^w = a_{\gamma,1} t_{1,\max}\$_1^w + \cdots + a_{\gamma,p} t_{p,\max}\$_p^w + \$_{\text{grav}}^w \quad 0 \le a_{\gamma,i} \le 1.$$

Thus, $\$_\gamma^w \in \text{NW}_{\text{avail}}$. Therefore, we conclude that NW_{avail} is convex.

Because NW_{avail} is convex, if the set of vertices V is contained in NW_{avail}, then $\text{conv}(V)$, the convex hull of V, is contained in NW_{avail}. Because NW_{req} is a set of polyhedra, $\text{NW}_{\text{req}} \subseteq \text{conv}(V)$. Thus, because $\text{conv}(V) \subseteq \text{NW}_{\text{avail}}$, $\text{NW}_{\text{req}} \subseteq \text{NW}_{\text{avail}}$. Therefore, $\text{NW}_{\text{req}} \subseteq \text{NW}_{\text{avail}} \Leftarrow V \subseteq \text{NW}_{\text{avail}}$. Q.E.D.□

[7]Note that for this proof and for the generation of the WFW, a collection of polyhedra can be equivalently replaced with the convex hull of the vertices of the polyhedra.

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APPENDIX II
COEFFICIENTS FOR EQUATION (19)

$$
\begin{aligned}
a_1 = &(F_y + mg) \\
&\times (\sin(\theta)(c_{j,x} - c_{i,x}) + \cos(\theta)(c_{j,y} - c_{i,y})) \qquad (32)\\
a_2 = &[(r_{i,y} - r_{j,y}) - \sin(\theta)(c_{i,x} - c_{j,x}) \\
&- \cos(\theta)(c_{i,y} - c_{j,y})]\, M - \ldots - (F_y + mg) \\
&\times [\cos(\theta)(c_{i,x}r_{i,y} - c_{j,x}r_{j,y} - c_{i,y}r_{i,x} + c_{j,y}r_{j,x}) - \ldots \\
&\quad - \sin(\theta)(c_{i,y}r_{i,y} - c_{j,y}r_{j,y} + c_{i,x}r_{i,x} - c_{j,x}r_{j,x})] + \ldots \\
&+ [F_x(r_{i,y} - \sin(\theta)c_{i,x} - \cos(\theta)c_{i,y}) - \ldots \\
&\quad - (F_y + mg)(r_{i,x} - \cos(\theta)c_{i,x} + \sin(\theta)c_{i,y})] \\
&\times (\sin(\theta)c_{j,x} + \cos(\theta)c_{j,y}) + \ldots \\
&+ [(F_y + mg)(r_{j,x} - \cos(\theta)c_{j,x} + \sin(\theta)c_{j,y}) - \ldots \\
&\quad - F_x(r_{j,y} - \sin(\theta)c_{j,x} - \cos(\theta)c_{j,y})] \\
&\times (\sin(\theta)c_{i,x} + \cos(\theta)c_{i,y}) \qquad (33)\\
a_3 = &(F_y + mg) \\
&\times [(\cos(\theta)(c_{i,x} - c_{j,x}) - \sin(\theta)(c_{i,y} - c_{j,y})] + \cdots \\
&+ F_x\,[(\sin(\theta)(c_{i,x} - c_{j,x}) + \cos(\theta)(c_{i,y} - c_{j,y})] \qquad (34)\\
a_4 = &[(r_{j,x} - r_{i,x}) - \cos(\theta)(c_{j,x} - c_{i,x}) \\
&+ \sin(\theta)(c_{j,y} - c_{i,y})]\, M + \ldots + F_x \\
&\times [\cos(\theta)(c_{i,x}r_{i,y} - c_{j,x}r_{j,y} - c_{i,y}r_{i,x} + c_{j,y}r_{j,x}) - \ldots \\
&\quad - \sin(\theta)(c_{i,y}r_{i,y} - c_{j,y}r_{j,y} + c_{i,x}r_{i,x} - c_{j,x}r_{j,x})] - \ldots \\
&- [F_x(r_{i,y} - \sin(\theta)c_{i,x} - \cos(\theta)c_{i,y}) - \ldots \\
&\quad - (F_y + mg)\,(r_{i,x} - \cos(\theta)c_{i,x} + \sin(\theta)c_{i,y})] \\
&\times (\cos(\theta)c_{j,x} - \sin(\theta)c_{j,y}) - \ldots \\
&- [(F_y + mg)\,(r_{j,x} - \cos(\theta)c_{j,x} + \sin(\theta)c_{j,y}) - \ldots \\
&\quad - F_x\,(r_{j,y} - \sin(\theta)c_{j,x} - \cos(\theta)c_{j,y})] \\
&\times (\cos(\theta)c_{i,x} - \sin(\theta)c_{i,y}) \qquad (35)\\
a_5 = &F_x\,(\cos(\theta)(c_{j,x} - c_{i,x}) - \sin(\theta)(c_{j,y} - c_{i,y})) \qquad (36)\\
a_6 = &(r_{i,x} - \cos(\theta)c_{i,x} + \sin(\theta)c_{i,y}) \\
&\times (r_{j,y} - \sin(\theta)c_{j,x} - \cos(\theta)c_{j,y})\, M - \ldots \\
&- (r_{i,y} - \sin(\theta)c_{i,x} - \cos(\theta)c_{i,y}) \\
&\times (r_{j,x} - \cos(\theta)c_{j,x} + \sin(\theta)c_{j,y})\, M + \ldots \\
&+ [F_x\,(r_{i,y} - \sin(\theta)c_{i,x} - \cos(\theta)c_{i,y}) - \ldots \\
&\quad - (F_y + mg)\,(r_{i,x} - \cos(\theta)c_{i,x} + \sin(\theta)c_{i,y})] * \ldots \\
&* [\cos(\theta)(c_{j,x}r_{j,y} - c_{j,y}r_{j,x}) \\
&\quad - \sin(\theta)(c_{j,y}r_{j,y} + c_{j,x}r_{j,x})] - \ldots \\
&- [F_x\,(r_{j,y} - \sin(\theta)c_{j,x} - \cos(\theta)c_{j,y}) - \ldots \\
&\quad - (F_y + mg)\,(r_{j,x} - \cos(\theta)c_{j,x} + \sin(\theta)c_{j,y})] * \ldots \\
&* [\cos(\theta)(c_{i,x}r_{i,y} - c_{i,y}r_{i,x}) \\
&\quad - \sin(\theta)(c_{i,y}r_{i,y} + c_{i,x}r_{i,x})]\,. \qquad (37)
\end{aligned}
$$

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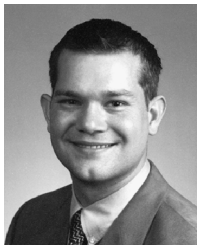

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